RESUME

PERSONAL



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PROFILE

Electronics and Instrumentation Engineering student with hands-on experience in Embedded Systems, IoT, PLC Automation, and Industrial Data Acquisition. Strong foundation in C Programming, Microcontrollers, Signal Processing, and Problem Solving. Currently building foundational knowledge in digital logic design, PWM techniques, electric drive/motor control, and FPGA-based design (VHDL/Verilog), alongside strong hands-on skills in serial communication protocols (UART/SPI/I2C). Seeking a Graduate Engineer Trainee role to contribute to Embedded Hardware, FPGA Design, and Power Electronics Development.

SKILL SET

- **Skill 1: Programming**
  - C
  - C++
  - Python basics
- **Skill 2: Embedded Systems**
  - STM32 basics
  - ESP32
  - UART
  - SPI
  - I2C, PWM Control (learning), FPGA Basics - VHDL/Verilog (learning),
  - Electric Drives & Motor Control (learning)
- **Skill 3: Industrial Automation**
  - PLC(ladder logic+HMI)
  - Siemens Technomatix
  - Industrial IoT
- **Skill 4: Software Tools**
  - Git
  - Node-Red
  - GenAI Tools

EDUCATION .

B.E. Electronics and Instrumentation Engineering
Kumaraguru College of Technology
Graduation: **2027**
CGPA: 8.0

Class 12: Bharatiya Vidya Bhavan **– 80%**

Class 10: Bharatiya Vidya Bhavan **– 100%**

ADDITIONAL QUALIFICATION .

Course Learning	Institution	Duration
●Embedded C	Coursera	1Months
●Wokwi,Proteus simulation	Kumaraguru College of Technology	1 Month
●AI Tools	be 10x.	24 hours
● Artificial Intelligence	United Latino Students Association.	1 Month
●Learning Industrial Automation.	Linked In Learning	1Month
●Embedded Microcontroller.	Kumaraguru College of Technology.	5Months

WORKING EXPERIENCE .

TANSAM (Siemens) — Intern(Duration: 1 Month)

- Worked on embedded systems and IoT applications
- Performed debugging, testing, calibration, and documentation
- Assisted in automation projects and prototype development
- Supported real-time monitoring implementation

Salzer Electronics — Intern(Duration: 15 Days)

- Learned industrial electronics workflows
- Assisted in testing and troubleshooting electronic systems
- Gained exposure to hardware components and quality control

AKS Transports — Intern(Duration: 1 Month)

- Worked on maintenance tracking and reporting systems
- Supported operational scheduling activities

PROJECTS

PROMO-X

- Collects PLC machine data using **Raspberry Pi** via profinet communication and sends it to a dashboard.
- Uses AI analytics for **anomaly detection** and **predictive maintenance** to reduce downtime.

Smart Vest - Design and Implementation of a Wireless IoT-Based Multi-Sensor Monitoring Platform

- Developed an **ESP32-based embedded system**
- Integrated multiple sensors for health monitoring
- Used **ESP32 communication** for wireless transmission

  Built real-time dashboard for monitoring

Wireless Temperature Data Transmission Using STM32

- Designed embedded firmware for sensor data collection
- Implemented wireless transmission system
- Performed testing and debugging

Portable Temperature Sensor Calibrator

- Designed a portable calibration device

  Improved sensor measurement accuracy

RFID based exam hall guidence system

- Developed RFID-based identification system
- Integrated hardware modules for automated hall allocation

Temperature-Based Fan Control System.

- Developed automatic fan/heater control system using temperature sensors with threshold-based control logic
- Implemented embedded logic for automation

OTHERS .

Strengths:

- Hands-on problem-solving ability
- Team Collaboration
- Quick learning
- Embedded Development

Computer literacy:

- Experienced in AI tools and programming
- IoT systems
- Data logging
- Dashboards using GenAI

Area of Interest

- Embedded Systems
- FPGA Design (VHDL/Verilog)
- Power Electronics & Electric Drives
- Internet of Things (IoT)
- Industrial Automation and PLC Systems
- Testing and Production Projects

Achievements /Awards :

- Selected as the intern at TANSAM, based on my work on the ResQLink project, demonstrating early trust in my technical skills and responsibility
- Successfully completed an industry-oriented internship at TANSAM,Chennai involving embedded systems, IoT, PLC automation, and real-time dashboards.
- I was shortlisted and selected as an intern at Salzer where I successfully completed a 15-day internship program. During this period, I gained practical exposure to technical workflows, industry practices, and professional work environments.
- I have successfully completed MicrochipTry - Basics of Developing Microcontroller Applications on Microchip University.
- Completed Introduction to Artificial Intelligence (Al) course by United Latino Students Association

Languages Known

- English
- German
- Telugu
- Tamil

Hobbies

- Listening to music
- Playing hockey
- Driving

SIGNATURE



NAME:Sanchith.R